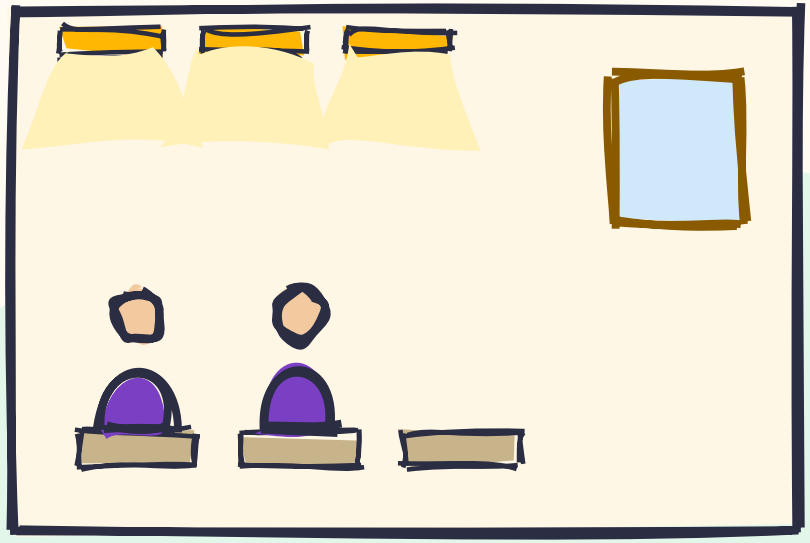
Young Innovators

Smart Classroom Energy Saver

Lights that switch on only when someone needs them

PIR + LDR sensing | AND logic | Relay switching

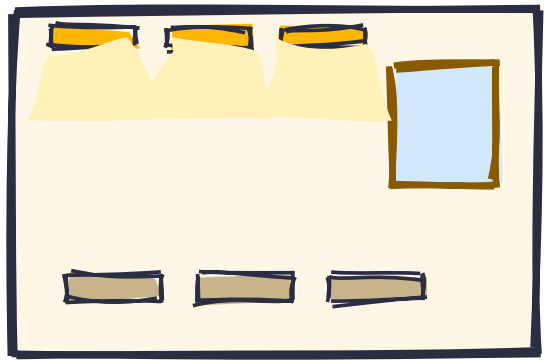


Did you know?
Lighting can be a big part of a school's electricity bill, and a lot of it lights empty rooms.

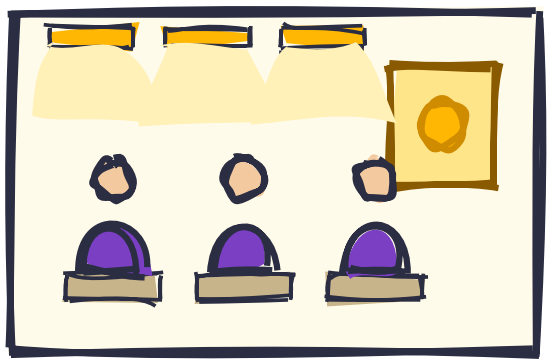
LEARN

Look around school...

Who is paying for the empty room?



Break time: nobody inside



Sunny afternoon: lights still on

Your mission

Design a controller that switches the light **ON** only when the room is **occupied** AND **too dark**. Then prove how much energy it saves.

Put up a sign?

Appoint a light monitor?

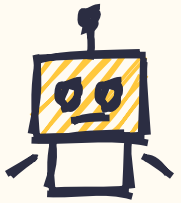
Let the room decide!

Learning aims

By the end of this session, I can…

1. Explain how a **PIR** sensor detects people and an **LDR** measures light
2. Combine two conditions with **AND** logic (and know when to use **OR**)
3. Switch a lamp safely through a **relay**
4. Show room status and **minutes saved** on an OLED
5. **Estimate** the energy and money our design saves in a year

Keywords
PIR · LDR · AND / OR · relay · millis() · watt · kWh · automation



LEARN

Watch and think

Video time!



How PIR Sensors Work: 3D Animated

Blue Butterfly · about 5 minutes · youtu.be/mXtQtpYg3k0

Watch for...

1. What does a **PIR** sensor detect?
2. Why are there two halves inside it?
3. Where could PIR sensors save energy in a school?

Scan to watch

After the video: Our classroom saver turns off lights in empty rooms. How long should it wait before switching off?

The science

Power, time and energy

Power (watts, W)

How fast a device uses energy. An LED tube light uses about **18 W**. An old fluorescent tube about **40 W**.

Energy (kWh)

Energy = power × time.
1000 W for 1 hour = **1 kWh** = the "1 unit" on your electricity bill.

Quick check

6 lights × 18 W for 2 hours = ?
_____ Wh = _____ kWh

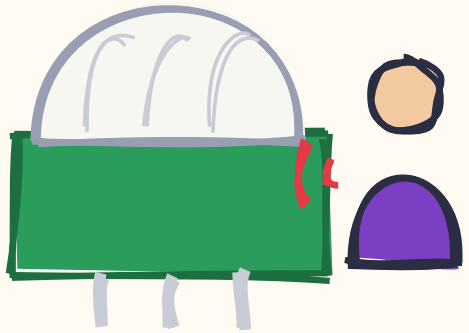


Much of Pakistan's electricity still comes from burning **fuel**. Every wasted unit costs money **and** adds pollution.

Answer
6 × 18 W × 2 h = 216 Wh = 0.216 kWh, every single day!

Sensor 1

PIR: it feels your body heat move



PIR HC-SR501

Pin	Connect / meaning
VCC	5V
OUT	HIGH (3.3 V) when motion
GND	Ground

How it works

Everything warm gives off **infrared**. The PIR has two IR detectors behind a **Fresnel lens** (the white dome). When a warm body moves from one zone to the other, the signals differ and OUT goes HIGH.

Fact file

Range about **7 m**, cone about **110°**. Two dials: **sensitivity** and **hold time** (about 3 s to 5 min). Needs about **1 minute** to settle after power-on.

The big catch

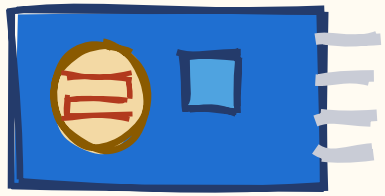
PIR sees **movement**, not people. A student sitting very still during a test can look like an **empty room**!

Think

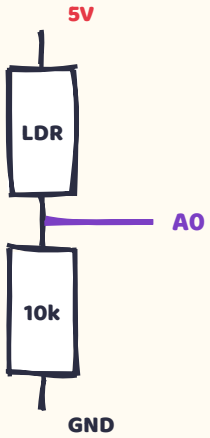
How should our code deal with a room full of very still students?

Sensor 2

LDR: how bright is the room?



LDR module



Inside the module: a **voltage divider**. More light, lower LDR resistance, so the voltage at A0 changes.

How it works

An **LDR** (light dependent resistor) has **less resistance in bright light**. The Uno reads the voltage as a number from **0 to 1023**.

Test	Your reading
Torch on it	_____
Normal classroom	_____
Hand covering it	_____

Check the direction!

Some modules give a **higher** number in the dark, some **lower**. Test yours first, then write your **DARK** threshold.

Tip

Pick DARK halfway between your 'normal classroom' and 'covered' numbers.

Decide

AND vs OR: which one saves energy?

Occupied?	Dark?	AND → lamp	OR → lamp
No	No	OFF	OFF
No	Yes	OFF	ON
Yes	No	OFF	ON
Yes	Yes	ON	ON

```
bool lampOn = occupied && dark; // AND
bool lampOn = occupied || dark; // OR
```

AND (&&)

Lamp is ON only when **both** are true. Empty room or bright room: lamp OFF. **This is our energy saver.**

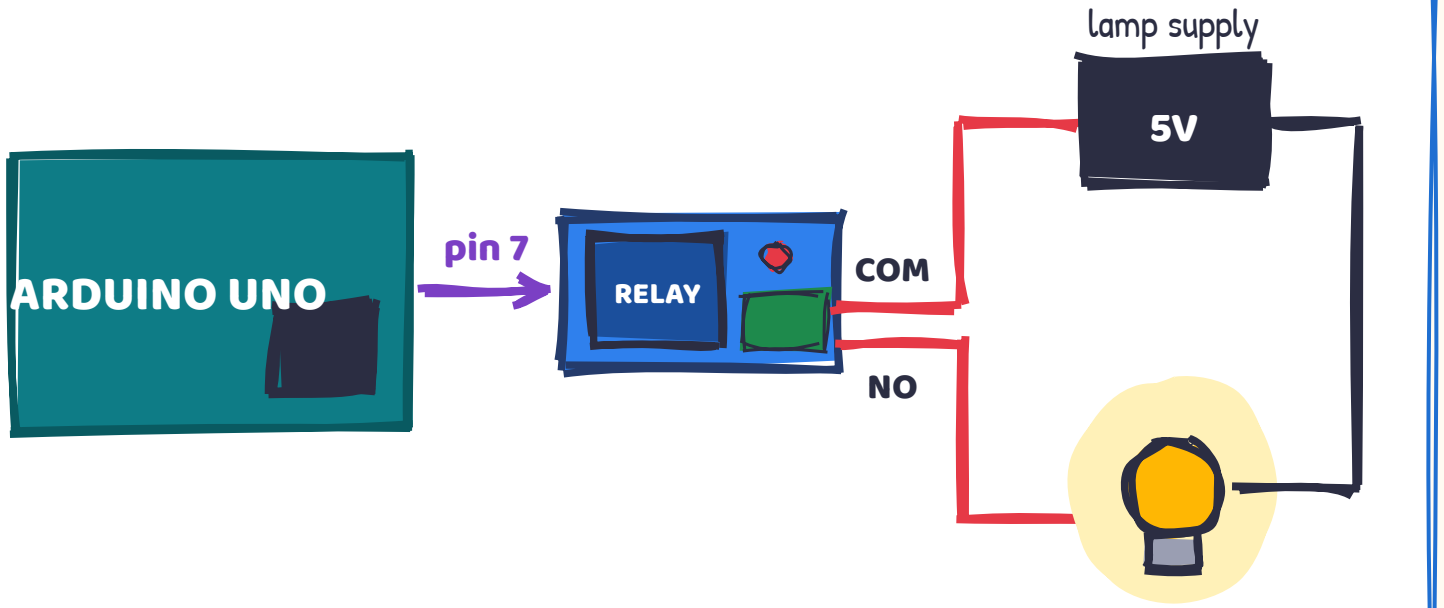
OR (||)

Lamp is ON when **either** is true. It would light an **empty** room at night. Great for a security light, bad for saving energy.

Count it: AND turns the lamp on in **1** of 4 cases. OR turns it on in **3** of 4.

Act

The relay: a switch flipped by a signal



Inside the relay

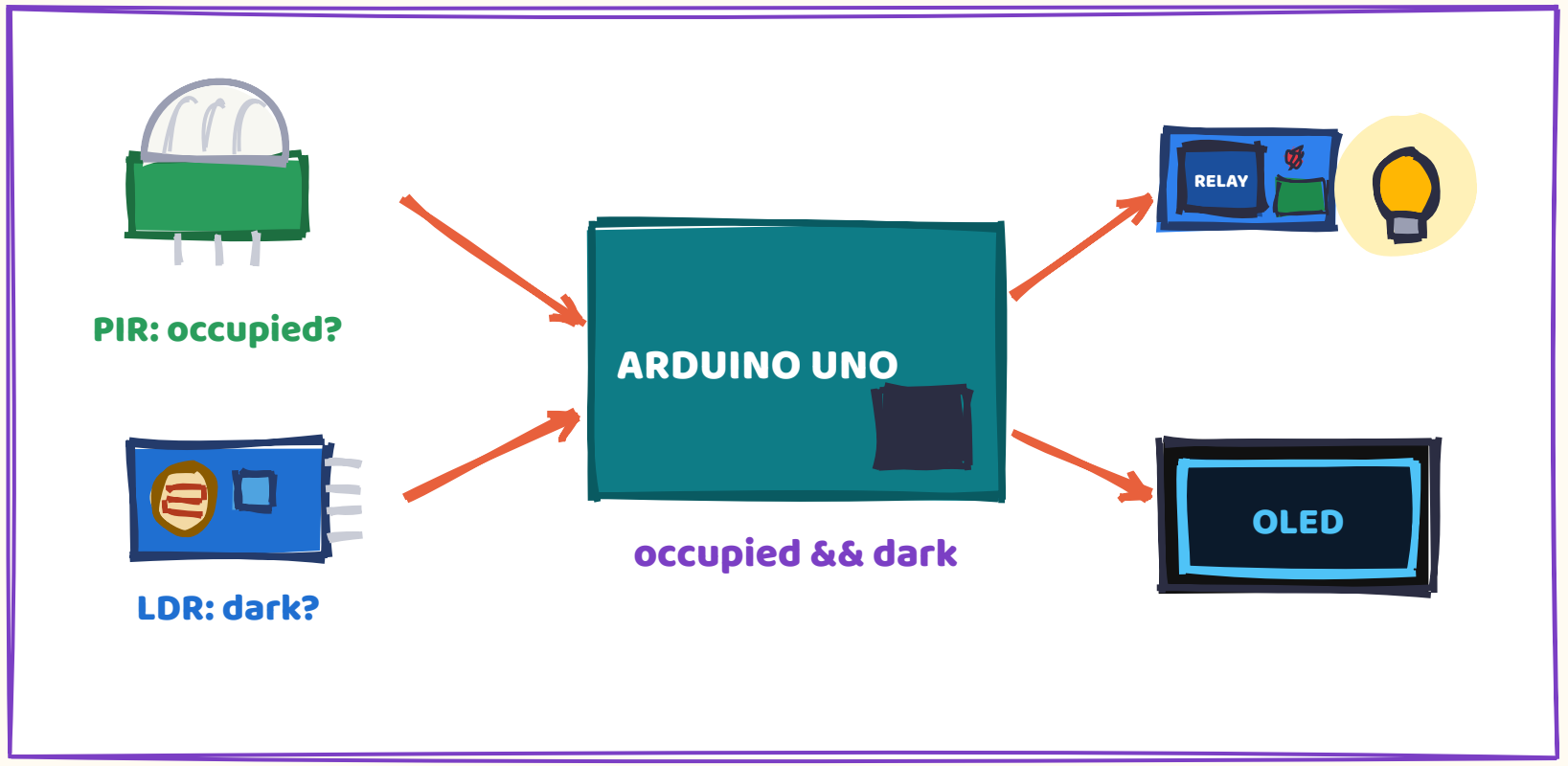
A tiny **electromagnet**. The Uno pin turns it on, it pulls a metal contact, and **click**: the lamp circuit closes. The two circuits never touch.

COM	Common: always used
NO	Normally Open: ON when relay clicks
NC	Normally Closed: opposite

Safety rule: we switch a **5 V** lamp only. The relay says 250 V on the box, but **mains wiring is for licensed electricians**, never for students.

The plan

Two sensors, one rule, two outputs




```
Room: OCCUPIED
Light: DARK
Lamp: ON
Saved: 37 min
```

What the OLED shows

Room status, light level, lamp state and a running count of minutes the lamp stayed OFF.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Audit our corridor (10 minutes)
2. Our energy saver kit (per group)
3. Wiring the energy saver
4. The decision (Arduino IDE)
5. The status screen
6. Does the logic hold up?
7. How much could the whole school save?

Be an energy detective

Audit our corridor (10 minutes)

Room	Lights on?	People?	Bright outside?	Wasted?
Room 1				
Room 2				
Room 3				
Corridor				

Rule of the audit

Wasted = lights ON while the room is **empty**, or ON while the room is already **bright**.

Count the lights in each room too. You will need this number for the **energy maths** at the end!

Real engineers

Every good design starts with real data, not guesses.

Build it

Our energy saver kit (per group)

Arduino Uno
The brain

PIR HC-SR501
Detects movement

LDR module
Measures light

1-channel relay
The electric switch

5 V LED lamp
Our 'classroom light'

OLED 0.96 in
Status screen

Also: breadboard, jumper wires, USB cable, laptop with **Arduino IDE 2** and the **Adafruit SSD1306** library installed.

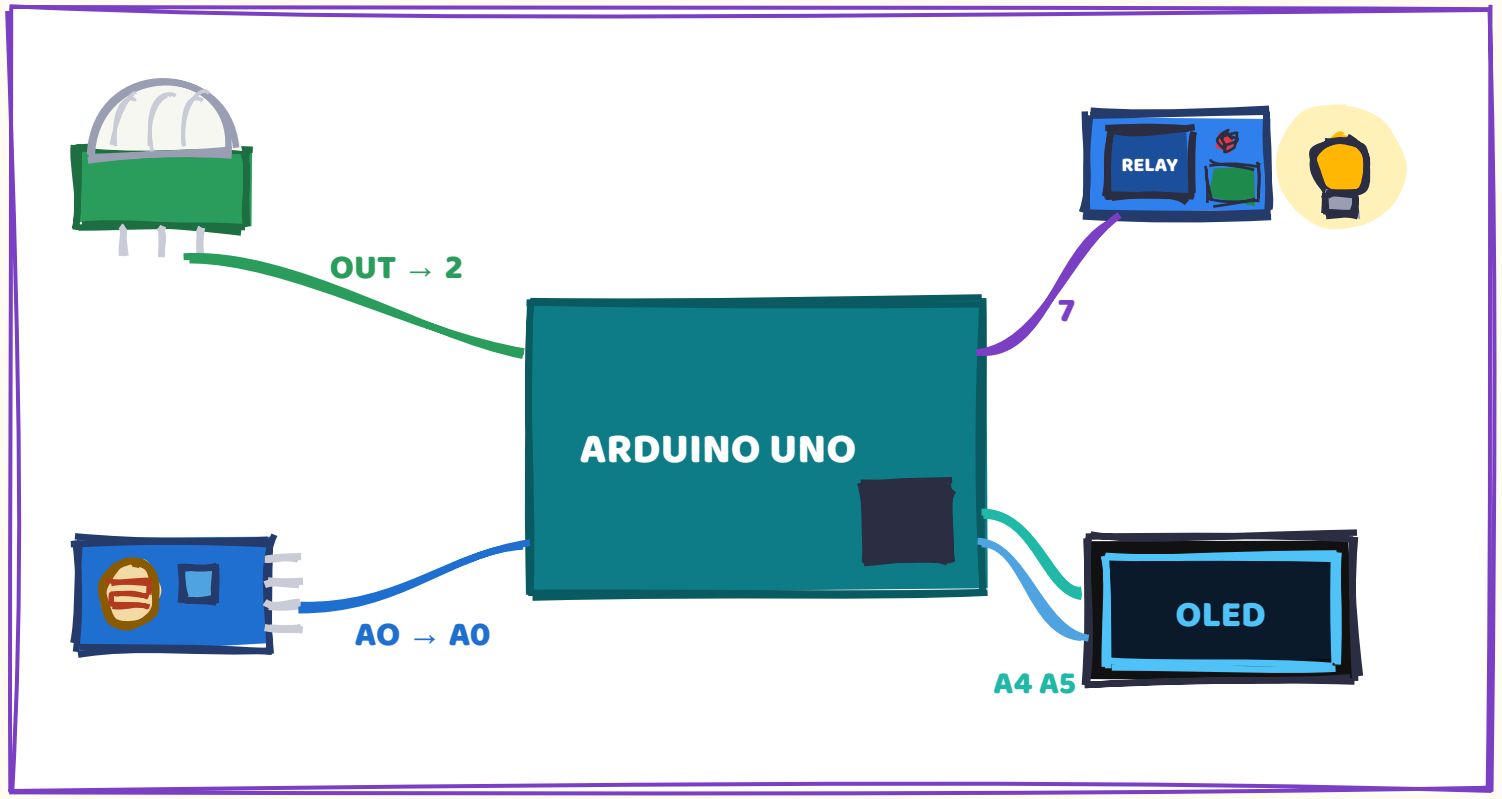
Try it first

Wokwi has a PIR, an LDR, a relay and an OLED. Test your logic before you wire!

MAKE

Build it

Wiring the energy saver



Part	Arduino
PIR VCC / GND	5V / GND
PIR OUT	pin 2
LDR VCC / GND	5V / GND
LDR AO	A0
Relay VCC / GND	5V / GND
Relay IN	pin 7
OLED SDA / SCL	A4 / A5
Lamp	5V supply via COM / NO

Active LOW?
Many relay boards switch ON with LOW. If the lamp works backwards, swap HIGH and LOW.

MAKE

Code it, part 1

The decision (Arduino IDE)

```
#include <Wire.h>
#include <Adafruit_SSD1306.h>
#define PIR   2
#define LDR   A0
#define RELAY 7
Adafruit_SSD1306 oled(128, 64, &Wire);

const int DARK = 600;        // from YOUR LDR test
unsigned long offMs = 0, lastTick = 0;

void setup() {
  pinMode(PIR, INPUT);
  pinMode(RELAY, OUTPUT);
  oled.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  delay(60000);              // let the PIR settle
  lastTick = millis();       // start the saving clock now
}

void loop() {
  unsigned long now = millis();
  bool occupied = digitalRead(PIR) == HIGH;
  bool dark = analogRead(LDR) > DARK;  // flip > if yours is reversed
  bool lampOn = occupied && dark;      // AND logic
  digitalWrite(RELAY, lampOn ? HIGH : LOW);
  if (!lampOn) offMs += now - lastTick;   // time saved
  lastTick = now;
  showStatus(occupied, dark, lampOn, offMs / 60000);
  delay(200);
}
```

Read the code

bool stores true or false.

&& is AND.

? : picks HIGH or LOW in one line.

millis() = milliseconds since start. We add up the time the lamp is OFF.

offMs / 60000 turns it into minutes.

Why add, not set?

offMs keeps growing all day. It is our energy-saving scoreboard!

MAKE

Code it, part 2

The status screen

```
void showStatus(bool occ, bool dark, bool on, unsigned long mins) {
  oled.clearDisplay();
  oled.setTextSize(1);
  oled.setTextColor(SSD1306_WHITE);
  oled.setCursor(0, 0);
  oled.print("Room:  ");  oled.println(occ  ? "OCCUPIED" : "EMPTY");
  oled.print("Light: ");  oled.println(dark ? "DARK" : "BRIGHT");
  oled.print("Lamp:  ");  oled.println(on   ? "ON" : "OFF");
  oled.setTextSize(2);
  oled.setCursor(0, 40);
  oled.print(mins);  oled.print(" min");   // minutes saved
  oled.display();
}
```



Nothing happens until display()! The OLED draws into memory first, then shows everything at once. No flicker.

Blank screen?
Check SDA on A4, SCL on A5, and try address 0x3D instead of 0x3C.

Test like an engineer

Does the logic hold up?

Test	Expect	Result
Walk in, lights off (cover LDR)	Lamp ON	pass / fail
Walk in, torch on LDR	Lamp OFF (bright)	pass / fail
Leave, stay out of view	Lamp OFF after hold time	pass / fail
Sit very still for 1 minute	?? Record what happens	pass / fail
Watch 5 minutes	Saved counter goes up	pass / fail

Debug checklist

Lamp always ON? Check the relay is not active LOW.
PIR always HIGH? Wait 60 s, and turn the hold-time dial down.
Wrong in bright light? Flip the > in the dark test.

Use **Serial.println()** to print the LDR number and PIR state every loop.

Prove it

How much could the whole school save?

Worked example

Lights per room	6 × 18 W = 108 W
Wasted time per day	2 hours
Energy per room per day	108 W × 2 h = 0.216 kWh
School days per year	× 200 = 43.2 kWh
Classrooms	× 40 = 1,728 kWh
Cost (example Rs 60 per unit)	≈ Rs 1,03,680 a year

Now use YOUR numbers

Lights per room (from your audit): ____

Watts per light: ____

Wasted hours per day: ____

Rooms in our school: ____

Rate on the school bill: Rs ____ per unit

Saving per year: Rs _____

Present it

Each group pitches its saving in 1 minute, like engineers pitching to the principal.

MAKE

Level up!

Energy saver challenges

Main challenge: 2-minute timeout

Fix the "still students" problem: keep the light on for 2 minutes after the **last** movement.

```
const unsigned long TIMEOUT = 120000;  // 2 min
unsigned long lastMotion = 0;
if (digitalRead(PIR) == HIGH) lastMotion = now;
bool occupied = (now - lastMotion) < TIMEOUT;
```

How it works

Every time the PIR sees movement, we **save the time** in lastMotion.
The room counts as occupied until **2 minutes** have passed with no movement at all.
Use **10000** (10 s) while testing, so you do not wait 2 minutes every time!

More challenges

- **kWh meter:** show saved energy in Wh, not minutes.
- **Manual override:** a button that forces the lamp ON for 10 min.
- **Dimmer:** use PWM on an LED to dim instead of switching off.

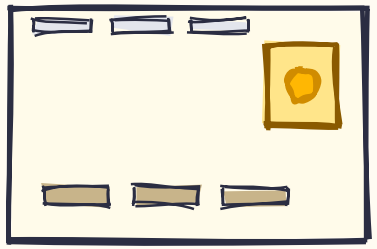
Boss level
Two lamps: the window row switches off first when the sun comes in.

Show what you know

What did we learn?

- [x] A PIR detects moving body heat, not still people
- [x] An LDR's resistance drops in bright light
- [x] AND (&&) needs both conditions; OR (||) needs either
- [x] A relay lets a small signal switch a separate circuit
- [x] Energy (kWh) = power (kW) × time (h)

At home
Find one light or fan at home that is often on with nobody there. Work out its kWh per month.



Exit ticket: a security light should come on at night when ANYONE walks past. AND or OR? Why?

Next: a Stepper Motor Drawing Plotter!